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Marc Pantel (Toulouse INP)

Subject: Application for PhD Position – Formal Engineering Methods for Digital Twin Development

Dear Marc Pantel,

I am applying for the PhD position on "Formal Engineering Methods for Digital Twin Development" with great enthusiasm. With a strong engineering background in Artificial Intelligence, distributed computing, and complex system modeling, I am highly motivated to contribute to the Artemis framework and tackle the unique challenges presented by this project.

My background in software engineering and safety-critical system design draws me strongly to formal methods. When developing autonomous cybersecurity platforms, ensuring system resilience required rigorous architectural constraints. I am highly motivated to extend formal verification techniques to the digital twin domain, ensuring that continuous data synchronization does not violate the strict safety properties required by aerospace and railway applications.

I am particularly drawn to this position because it bridges rigorous research with practical industrial challenges within the EDT program. Please find attached my resume and academic transcripts, which detail my technical competencies and engineering background. I welcome the opportunity to discuss my application further.

Sincerely,

Ismail Aissaoui